

Sending

27-09-2025

* Introduction to Automation

=> Automation is Software testing is the process of using specialized tools or scripts to perform tests on Software automatically instead of doing them manually. The goal is to save time, increase accuracy, and run repetitive tests efficiently.

Example:- Manual Testing Scenario, username, password.

2. Test Scripting Languages:-

=> Test Scripting languages are programming languages used to write automation scripts for scripts testing. They instruct tools on how to perform actions, verify results, and log outcomes.

Common Languages:-

- => python - Easy to learn, widely used in Selenium & PyTest.
- => Java - popular with Selenium WebDriver & TestNG Frameworks.
- => JavaScript - Used with tools like Cypress & Protractor.
- => C# - used in automation with Selenium & Visual Studio.

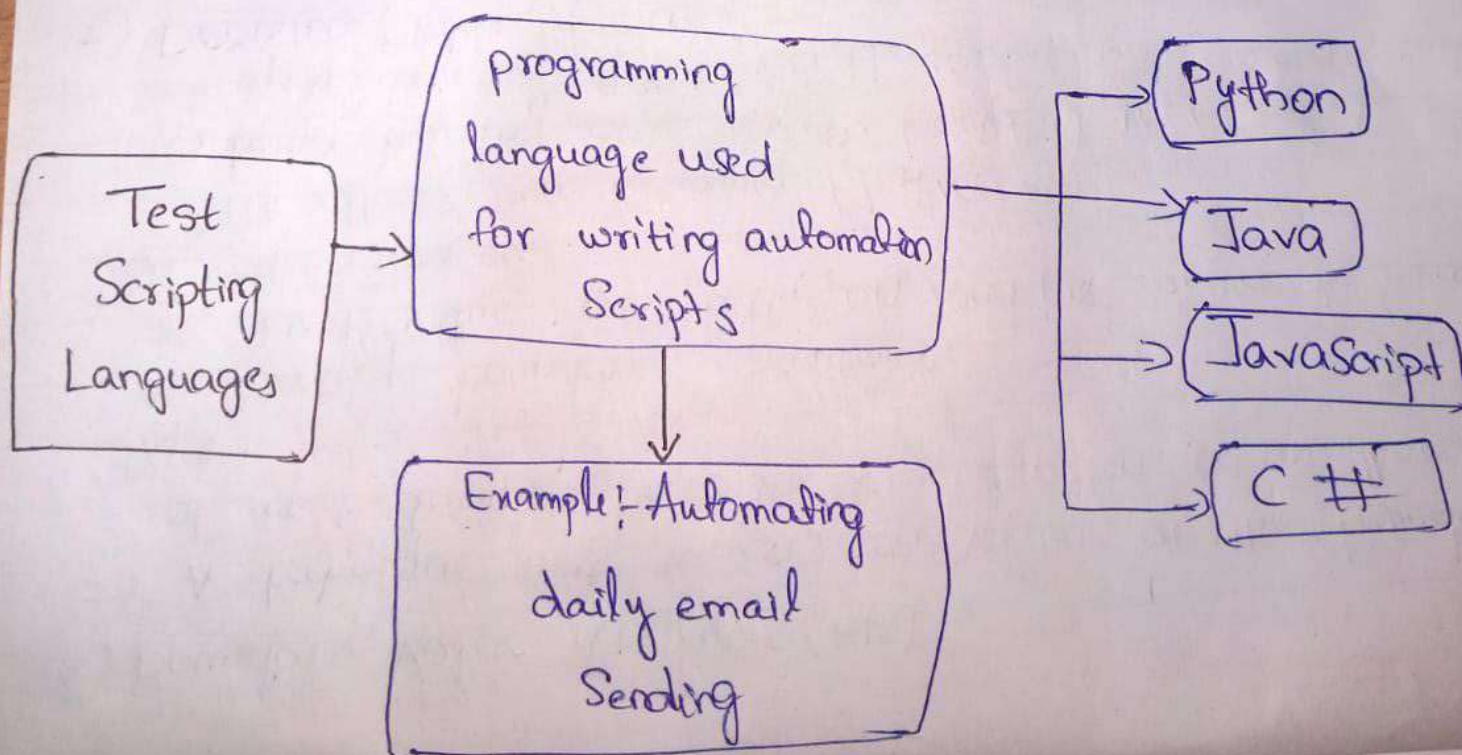
Ex:- Using python, you can write a script that connects to your email, drafts and selects send messages automatically.

- => In testing, python (or) Java scripts can automate filling forms on websites (or) checking data in applications repeatedly.
- => Test Scripting languages are the backbone of automation
 - without them, automation tools cannot interact with applications or execute tests. choosing the right language (python, Java, JavaScript, C#) depends on the tool, projects needs, and team expertise.

MANUAL



AUTOMATED



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3. Automation framework Selection :

=> An automation framework is a set of guidelines, standards, and tools that make automation testing more organized and maintainable. Frameworks help reduce code duplication and improve test efficiency.

Common Types:-

1. Linear Scripting framework - Simple record-and-playback scripts (good for small projects)
2. Modular Testing framework - Breaks tests into reusable modules
3. Data-Driven framework - Tests run with multiple sets of input data.
4. Keyword-Driven framework - Uses keywords to represent actions, easy for non-coders.
5. Hybrid framework - Combines multiple framework for flexibility.

4. Continuous Integration (CI) Tools.

=> Continuous Integration (CI) tools are software that automatically build, test, and deploy applications whenever changes are made to the code. CI ensures early detection of bugs and smooth integration of new code.

popular CI Tools:-

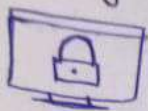
- => Jenkins - open-source, widely used for automation pipelines.
- => Github Actions - CI/CD integrated with Github.
- => Gitlab CI/CD - Automation directly in Gitlab
- => CircleCI, Travis CI - cloud-based CI Tools.

=> Ensures faster feedback, reduces errors, and maintains consistent, reliable software delivery.

AUTOMATION TESTING REAL-WORLD EXAMPLES

MANUAL TESTING

Login functionality



Testers enter UNK PASS every time a new build is released, clicks login verifies if the user is redirected correctly.

E-commerce checkout



Tester manually adds item to cart, applies discounts, enters payment details, and verifies if the order is placed correctly.

Data Entry & validation



Enter thousands of records in a database manually and check if they are saved correctly.

Regression Testing



After each update, testers check all previously working features manually to ensure nothing broke.

AUTOMATION TESTING

Automation Testing



using Selenium, a script enters the UNK PASS, clicks login checks if the user lands on the homepage automatically.

Automation Testing



A script can automatically.

1. Add items to Cart
2. Apply coupon codes
3. fill payment into
4. verify order confirmation ✓

Automation Testing



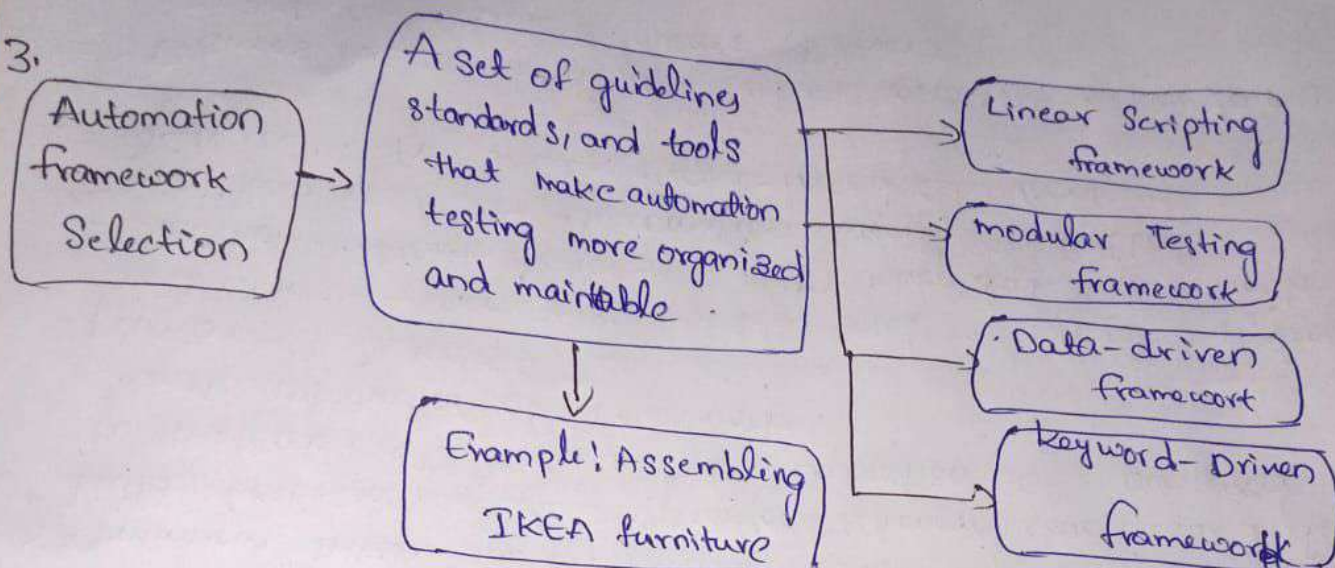
use a script to read data from Excel or CSV and insert it into the application automatically, then verify each entry.

Automation Testing

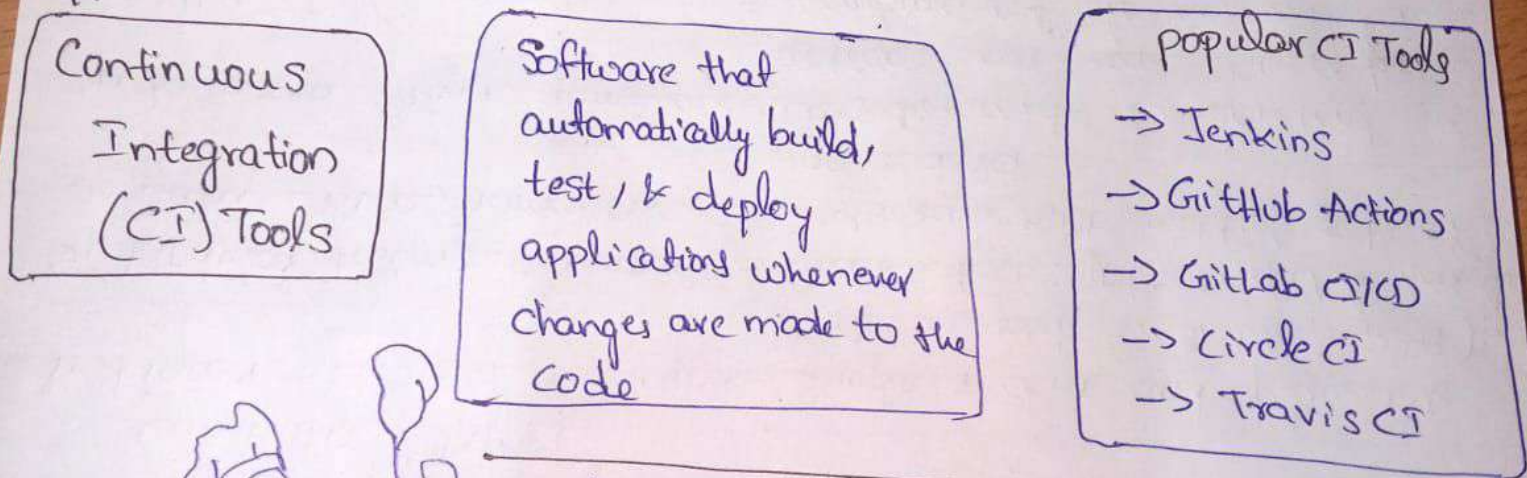


Tools like Appium can simulate multiple users interacting with the app on different devices and OS versions automatically.

3.



4.



Example! A team of chefs cooking in a large kitchen. Every time a new recipe is added (Code Commit), the CI tool automatically checks all ingredients (tests), Cooks a Sample